Boundary-Layer Transfer and Bounded Sharp Saint–Venant (Route δ, Plan 2, Building Block BoundaryLayerTransfer)

Abstract

We close the route-δ chain by transferring the metric-trace estimate at the inner level $\widehat{\rho}$ back to the original set Ω. The upstream weighted metric trace ([5]) supplies a radius $\widehat{\rho} \in [\rho_\delta - C_*\delta_T, \rho_\delta]$, with $\rho_\delta = 1 - \kappa\sqrt{\delta_T}$, such that $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \le C_*\, \delta_T(\Omega)$. The boundary-layer volume $|\Omega \setminus E_{\widehat{\rho}}| = \omega_n(1-\widehat{\rho}^n)$ is $O(\sqrt{\delta_T})$; the elementary superlevel-transfer estimate of [1] (Lemma 8.3) gives $\mathcal{A}(\Omega) \le \mathcal{A}(E_{\widehat{\rho}}) + C_n\sqrt{\delta_T}$. Squaring with $(a+b)^2 \le 2a^2 + 2b^2$ doubles the constant on δ_T but preserves the sharp exponent 2. We obtain the bounded sharp Saint–Venant inequality $\mathcal{A}(\Omega)^2 \le C(n, R, \rho_*)\, \delta_T(\Omega)$ for all open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ inside an explicit smallness window $\delta_T(\Omega) \le \delta_0(n, R, \rho_*)$. Outside that window the inequality is trivial after enlarging C via the FMP quantitative isoperimetric inequality applied to Ω. Every constant is polynomial in $1/\rho_*$, R, n and the constants of the upstream building blocks.

1 Statement of the theorem

Throughout, $n \ge 2$, $R \ge 2$, $\rho_* \in (0, 1)$ are fixed. $\Omega \subset \mathbb{R}^n$ denotes a bounded open set with $|\Omega| = \omega_n$ and $\Omega \subset B_R$. The torsion functional is normalised by $\delta_T(\Omega) := E(\Omega) - E(B_1) \ge 0$. The Fraenkel asymmetry is

$$\mathcal{A}(\Omega) := \inf\left\{ \frac{|\Omega\Delta(z + B_1)|}{\omega_n} \ :\ z \in \mathbb{R}^n \right\} \in [0, 2).$$

For $\rho \in [\rho_*, 1]$ we write $E_\rho = \{u_\Omega > t(\rho)\}$ where $|E_\rho| = \omega_n\rho^n$ and u_Ω is the torsion function of Ω (see [1]). The normalised level set $F_\rho := \rho^{-1}E_\rho \in \mathcal{X}$, where $\mathcal{X}$ is the metric quotient of finite-measure sets by translations equipped with

$$d_{\mathcal{X}}([A], [C]) := \inf_{a\in\mathbb{R}^n} |A\Delta(C + a)|.$$

Theorem 1.1 (Bounded sharp Saint–Venant; building block BoundaryLayerTransfer)**.** *There exist $C = C(n, R, \rho_*) > 0$ and $\delta_0 = \delta_0(n, R, \rho_*) > 0$ such that every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ satisfies*

$$\mathcal{A}(\Omega)^2 \le C(n, R, \rho_*)\, \delta_T(\Omega). \tag{1}$$

The constant $C(n, R, \rho_*)$ is polynomial in $1/\rho_*$, R, n and the constants of [3], [4], [5]. The exponent 2 is sharp.

The proof occupies §2–§7: the small-deficit regime $\delta_T(\Omega) \le \delta_0$ is closed by the metric trace, the elementary superlevel transfer, and the squaring identity; the complementary regime $\delta_T(\Omega) \ge \delta_0$ is closed by the trivial bound $\mathcal{A}(\Omega) \le 2$ after absorbing $4/\delta_0$ into C.

2 The metric-trace input

We record, in the form needed below, the output of the weighted metric trace lemma assembled in [5] from [4] and [3]. Set

$$\rho_\delta := 1 - \kappa\sqrt{\delta_T(\Omega)}, \tag{2}$$

where $\kappa = \kappa(n, \rho_*) > 0$ is the profile-gap constant fixed in [4, §5]; equivalently ρ_δ is the largest radius for which the profile-gap lower bound on the measure $d\mu(\rho) = (-t'(\rho))\, d\rho$ holds (cf. [1, (5.1)]).

Theorem 2.1 (Metric trace at $\widehat{\rho}$; [5])**.** *There exist $C_* = C_*(n, R, \rho_*) > 0$ and $\delta_1 = \delta_1(n, R, \rho_*) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \le \delta_1$, there exists*

$$\widehat{\rho} = \widehat{\rho}(\Omega) \in [\rho_\delta - C_*\delta_T(\Omega),\ \rho_\delta] \tag{3}$$

such that

$$d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \le C_*\, \delta_T(\Omega). \tag{4}$$

Reference. This is the conclusion of [5, Theorem on weighted metric trace], combining the integrated metric-derivative bound [5, (4.2)] with the profile-gap measure [1, (5.1)] and the metric Sobolev trace lemma [5, Lemma 3.1]. No additional input is needed here. □

Remark 2.2. By the definition of $d_{\mathcal{X}}$ and the volume normalisation $|E_{\widehat{\rho}}| = \omega_n\widehat{\rho}^n$, there exists $z = z(\widehat{\rho}) \in \mathbb{R}^n$ such that $|E_{\widehat{\rho}}\Delta(z + B_{\widehat{\rho}})| \le \widehat{\rho}^n d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)$. Dividing by $|E_{\widehat{\rho}}| = \omega_n\widehat{\rho}^n$,

$$\mathcal{A}(E_{\widehat{\rho}}) \le \omega_n^{-1} d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1), \qquad \mathcal{A}(E_{\widehat{\rho}})^2 \le \omega_n^{-2} C_*\delta_T(\Omega). \tag{5}$$

Here $\mathcal{A}(E_{\widehat{\rho}})$ is the Fraenkel asymmetry of $E_{\widehat{\rho}}$ relative to balls of *its own volume* $\omega_n\widehat{\rho}^n$, not of ω_n.

3 Volume of the removed boundary layer

Lemma 3.1 (Boundary-layer volume)**.** *Let $\widehat{\rho}$ be as in Theorem 2.1. There is $C_n > 0$ depending only on n such that, for $\delta_T(\Omega) \le \delta_2(n, \rho_*)$ small,*

$$|\Omega \setminus E_{\widehat{\rho}}| = \omega_n\big(1 - \widehat{\rho}^n\big) \le C_n\sqrt{\delta_T(\Omega)}. \tag{6}$$

Proof. From (2)–(3),

$$\widehat{\rho} \ge \rho_\delta - C_*\delta_T(\Omega) = 1 - \kappa\sqrt{\delta_T(\Omega)} - C_*\delta_T(\Omega). \tag{7}$$

The map $\rho \mapsto 1 - \rho^n$ is C^1 with $\frac{d}{d\rho}(1 - \rho^n) = -n\rho^{n-1}$, so by the mean-value theorem applied on $[\widehat{\rho}, 1] \subset [\rho_*, 1]$,

$$1 - \widehat{\rho}^n \le n(1 - \widehat{\rho}) \le n\big(\kappa\sqrt{\delta_T(\Omega)} + C_*\delta_T(\Omega)\big).$$

For $\delta_T(\Omega) \le 1$, the second summand is bounded by $C_*\sqrt{\delta_T(\Omega)}$, giving (6) with $C_n := \omega_n n(\kappa + C_*)$. Choosing $\delta_2 := \min\{1, \delta_1\}$ ensures both (3) and $\delta_T(\Omega) \le 1$. □

4 The asymmetry transfer lemma

We now prove the elementary superlevel transfer inequality ([1], Lemma 8.3). The point is that the asymmetry of Ω at scale $|\Omega| = \omega_n$ is dominated by the asymmetry of any nested superlevel E_t at its own scale $|E_t| = \omega_n\rho^n$ plus a volume-defect correction; the ball-scale correction $|B_{\omega_n} \setminus B_s|$ is controlled by the same defect by elementary monotonicity.

Throughout this section, Ω has $|\Omega| = L = \omega_n$, $E_t \subset \Omega$ is open with $|E_t| = s$ and $\eta := L - s \in [0, L]$. We write B_r for the open ball of radius r centred at the origin and recall that the volume of B_r is $\omega_n r^n$, so the ball of volume s is B_{r_s} with $r_s := (s/\omega_n)^{1/n}$, and the ball of volume L is B_1.

Lemma 4.1 (Superlevel asymmetry transfer; cf. [1] Lemma 8.3)**.** *Let Ω, E_t be as above with $E_t \subset \Omega$ and $\eta = |\Omega| - |E_t|$. Then*

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_t) + \frac{2\eta}{L}. \tag{8}$$

Equivalently, since $L = \omega_n$,

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_t) + \frac{2}{\omega_n}\,|\Omega \setminus E_t|. \tag{9}$$

Proof. Pick $\varepsilon \in (0, L)$ and choose $z \in \mathbb{R}^n$ such that

$$|E_t \Delta (z + B_{r_s})| \le s\big(\mathcal{A}(E_t) + \varepsilon\big). \tag{10}$$

We compare Ω to the ball $z + B_1$ of volume L. The triangle inequality for symmetric differences gives, since $z + B_{r_s} \subset z + B_1$ (because $r_s \le 1$),

$$|\Omega \Delta (z + B_1)| \le |\Omega \Delta E_t| + |E_t \Delta (z + B_{r_s})| + |(z + B_{r_s}) \Delta (z + B_1)|. \tag{11}$$

We bound each summand.

First summand. Because $E_t \subset \Omega$, $\Omega \Delta E_t = \Omega \setminus E_t$, hence

$$|\Omega \Delta E_t| = \eta. \tag{12}$$

Second summand. (10) gives

$$|E_t \Delta (z + B_{r_s})| \le s\big(\mathcal{A}(E_t) + \varepsilon\big). \tag{13}$$

Third summand. Since $z + B_{r_s} \subset z + B_1$,

$$|(z + B_{r_s}) \Delta (z + B_1)| = |B_1 \setminus B_{r_s}| = L - s = \eta. \tag{14}$$

Combining (11)–(14) and dividing by L,

$$\frac{|\Omega \Delta (z + B_1)|}{L} \le \frac{2\eta}{L} + \frac{s}{L}\big(\mathcal{A}(E_t) + \varepsilon\big) \le \frac{2\eta}{L} + \mathcal{A}(E_t) + \varepsilon, \tag{15}$$

using $s \le L$. Since $\mathcal{A}(\Omega) \le L^{-1}|\Omega \Delta (z + B_1)|$ and $\varepsilon \in (0, L)$ is arbitrary, (8) follows. □

Remark 4.2 (On the ball-scale correction)**.** The triangle inequality (11) contains two distinct volume-defect terms: $|\Omega \Delta E_t| = \eta$ (the boundary layer of Ω outside E_t) and $|B_1 \setminus B_{r_s}| = \eta$ (the "ball-scale" correction). They add to give $2\eta/L$. There is no $|B_s|/|B_L|$ rescaling to worry about, because the symmetric difference $|E_t \Delta (z + B_{r_s})|$ is measured at the natural scale s of E_t, and the cross term to scale L is precisely $|B_1 \setminus B_{r_s}|$. This is the same combinatorial identity used in [1, Lemma 8.3] and in the Cicalese–Leonardi L^1-rearrangement bookkeeping [10].

Corollary 4.3 (Transfer at the level $\widehat{\rho}$)**.** *Let $\Omega \subset B_R$ with $|\Omega| = \omega_n$, $\delta_T(\Omega) \le \delta_2$, and let $\widehat{\rho}$ be as in Theorem 2.1. Then*

$$\mathcal{A}(\Omega) \le \mathcal{A}(E_{\widehat{\rho}}) + \frac{2C_n}{\omega_n}\sqrt{\delta_T(\Omega)}, \tag{16}$$

where C_n is the constant of Lemma 3.1.

Proof. $E_{\widehat{\rho}} \subset \Omega$ by definition, and $|\Omega \setminus E_{\widehat{\rho}}| \le C_n\sqrt{\delta_T(\Omega)}$ by Lemma 3.1. Apply (9) with $E_t = E_{\widehat{\rho}}$. □

5 Squaring preserves the exponent 2

We combine Corollary 4.3 with (5) and the elementary inequality $(a+b)^2 \le 2a^2 + 2b^2$.

Theorem 5.1 (Small-δ_T bound)**.** *There exist $C_3 = C_3(n, R, \rho_*) > 0$ and $\delta_0 = \delta_0(n, R, \rho_*) \in (0, \delta_2]$ such that every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \le \delta_0$ satisfies*

$$\mathcal{A}(\Omega)^2 \le C_3(n, R, \rho_*)\, \delta_T(\Omega). \tag{17}$$

Proof. Take $\delta_T(\Omega) \le \delta_2$ so that Corollary 4.3 applies. Set $a := \mathcal{A}(E_{\widehat{\rho}})$, $b := \frac{2C_n}{\omega_n}\sqrt{\delta_T(\Omega)}$. Then (16) reads $\mathcal{A}(\Omega) \le a + b$. Squaring,

$$\mathcal{A}(\Omega)^2 \le 2a^2 + 2b^2 = 2\mathcal{A}(E_{\widehat{\rho}})^2 + \frac{8C_n^2}{\omega_n^2}\delta_T(\Omega). \tag{18}$$

By (5), $\mathcal{A}(E_{\widehat{\rho}})^2 \le \omega_n^{-2} C_* \delta_T(\Omega)$. Substituting,

$$\mathcal{A}(\Omega)^2 \le \frac{2C_* + 8C_n^2}{\omega_n^2}\, \delta_T(\Omega) =: C_3\, \delta_T(\Omega). \tag{19}$$

This proves (17) with $\delta_0 := \delta_2$. The exponent 2 is preserved because $(a+b)^2 \le 2a^2 + 2b^2$ *doubles the constant* but does not raise either summand to a higher power. $\square$

Remark 5.2 (The factor of 2)**.** The identity $(a+b)^2 = a^2 + 2ab + b^2 \le 2(a^2 + b^2)$, valid by $2ab \le a^2 + b^2$, is sharp up to the factor of 2 (saturated when $a = b$). Replacing it by Young's inequality with a parameter $\theta \in (0,1)$, $(a+b)^2 \le (1+\theta^{-1})a^2 + (1+\theta)b^2$, allows the prefactor on $\mathcal{A}(E_{\widehat{\rho}})^2$ to be made arbitrarily close to 1, at the cost of inflating the coefficient on δ_T. This does not affect the exponent, only the constant C_3. For the bounded sharp Saint–Venant inequality the cleanest choice is $\theta = 1$, giving (19).

6 Constants chain

The constant $C_3 = C_3(n, R, \rho_*)$ produced by Theorem 5.1 unwinds as

$$C_3 = \frac{2C_* + 8C_n^2}{\omega_n^2}, \tag{20}$$

where the inputs are:

- $C_* = C_*(n, R, \rho_*)$: the metric trace constant of Theorem 2.1, supplied by [5, Theorem on weighted metric trace]. By the analysis in [4, §7], C_* is polynomial in $(1/\rho_*, R, n)$ and linear in the centroid-bound constant $C_F = C_F(n, R, \rho_*)$ of [3].
- $C_n = \omega_n n(\kappa + C_*)$ from Lemma 3.1, where $\kappa = \kappa(n, \rho_*)$ is the profile-gap constant of [4, §5].
- $\omega_n = |B_1|$ is purely dimensional.

In particular, C_3 is polynomial in $(1/\rho_*, R, n, C_F)$ and *tracks linearly* in the new structural input C_F from the centroid bound. Specialising to $R = R_*(n)$ (the bounded reduction radius) and $\rho_* = 1/2$ as in [5], C_3 depends only on n.

Remark 6.1 (Sharpness of the dependence)**.** The polynomial dependence on $1/\rho_*$ enters through (i) the profile-gap constant in [1, (5.1)]; (ii) the Cauchy–Schwarz bookkeeping in the metric derivative bound [5, (2.4),(4.2)]; and (iii) the oscillation-index input to the homothetic velocity defect [4]. None of these can be improved beyond polynomial without altering the upstream construction.

7 Smallness regime parametrisation and the global bound

Theorem 5.1 closes the small-δ_T regime. For the complementary regime $\delta_T(\Omega) \geq \delta_0$, the trivial bound $\mathcal{A}(\Omega) < 2$ suffices.

Lemma 7.1 (Large-δ_T trivial bound)**.** *For every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \geq \delta_0(n, R, \rho_*)$,*

$$\mathcal{A}(\Omega)^2 \leq \frac{4}{\delta_0(n, R, \rho_*)}\, \delta_T(\Omega). \tag{21}$$

Proof. $\mathcal{A}(\Omega) < 2$ by definition of the Fraenkel asymmetry, so $\mathcal{A}(\Omega)^2 < 4$. By assumption, $1 \leq \delta_T(\Omega)/\delta_0$, hence $\mathcal{A}(\Omega)^2 \leq 4 \leq 4\delta_T(\Omega)/\delta_0$. □

Remark 7.2 (Alternative via FMP)**.** A second proof of (21) uses the FMP quantitative isoperimetric/Saint–Venant inequality [9, 10] applied to Ω itself: $\mathcal{A}(\Omega)^4 \leq C_{\mathrm{FMP}}\delta_T(\Omega)$, which on $\delta_T(\Omega) \geq \delta_0$ implies a quadratic bound with constant $C'_{\mathrm{FMP}}/\delta_0^{1/2}$. The bound via $\mathcal{A} < 2$ is sharper for our purposes.

Proof of Theorem 1.1. For $\delta_T(\Omega) \leq \delta_0$, (1) holds with $C(n, R, \rho_*) := C_3(n, R, \rho_*)$ by Theorem 5.1. For $\delta_T(\Omega) \geq \delta_0$, Lemma 7.1 gives the same bound with $C(n, R, \rho_*) := 4/\delta_0(n, R, \rho_*)$. Taking

$$C(n, R, \rho_*) := \max\left\{ C_3(n, R, \rho_*), \frac{4}{\delta_0(n, R, \rho_*)} \right\} \tag{22}$$

proves (1) on the whole admissible class.

Sharpness of the exponent 2 is the standard ellipsoidal saturation of [11, Introduction]: along the family of volume-preserving ellipsoidal perturbations Ω_ε of B_1, $\mathcal{A}(\Omega_\varepsilon) \sim \varepsilon$ and $\delta_T(\Omega_\varepsilon) \sim \varepsilon^2$ to second order, so any bound $\mathcal{A}(\Omega)^p \leq C\delta_T(\Omega)$ requires $p \geq 2$. □

Remark 7.3 (How this slots into GlobalAssembly)**.** Theorem 1.1 is precisely the input to the bounded reduction step of [7, Theorem 5.1], specialised to $R = R_*(n)$. Together with the Kohler–Jobin transfer of [8, Theorem 6.1], it yields the sharp quantitative Faber–Krahn inequality

$$\big(|\Omega|/|B_1|\big)^{2/n}\big(\lambda_1(\Omega) - \lambda_1(B^*)\big) \geq c_{\mathrm{FK}}(n)\, \mathcal{A}(\Omega)^2,$$

as assembled in [6, §6].

Remark 7.4 (Where this block sits in the Plan 2 chain)**.** The boundary-layer transfer is the last *geometric* step in the route-δ chain. Upstream, [1] provides the exact deficit identity, [2] the metric first variation, [3] the H^1-in-ρ centroid bound, [4] the integrated profile-gap (B^2-int) inequality, and [5] the metric trace at $\widehat{\rho}$. The present block does *not* use any pointwise (R)-type input, only the metric trace (4) and the elementary triangle inequality of §4. In particular, no $C^{1,\gamma}$ regularity of $\partial\Omega$ is required; the proof is purely finite-perimeter.

References

[1] Plan 2 building block LevelSetIdentity. Exact deficit identity for finite-perimeter superlevel sets; Lemmas 8.1–8.3 (profile-gap measure, superlevel replacement).

[2] Plan 2 building block MetricFramework. Metric space $\mathcal{X}$, metric first variation, per-ρ velocity defect.

[3] Plan 2 building block CentroidBound. H^1-in-ρ Fraenkel-centroid bound (the new structural input); constant $C_F = C_F(n, R, \rho_*)$.

[4] Plan 2 building block SpaceTimeIdentity. Space-time Π identity; integrated (B^2-int) inequality via slicing-then-squaring; profile-gap measure constant κ.

[5] Plan 2 building block WeightedMetricTrace. Integrated action $\to$ pointwise endpoint at $\widehat{\rho}$; metric trace estimate $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)^2 \leq C_* \delta_T$ with $\widehat{\rho} \in [\rho_\delta - C_* \delta_T, \rho_\delta]$.

[6] Plan 2 building block GlobalAssembly. Bounded reduction + Kohler–Jobin transfer; final sharp Faber–Krahn inequality.

[7] `Final/BoundedReduction.tex`. Bounded reduction theorem at $R = R_*(n)$.

[8] `Final/KohlerJobinTransfer.tex`. Kohler–Jobin transfer from sharp Saint–Venant to sharp Faber–Krahn.

[9] N. Fusco, F. Maggi, A. Pratelli, *The sharp quantitative isoperimetric inequality*, Ann. of Math. **168** (2008), 941–980.

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[11] L. Brasco, G. De Philippis, B. Velichkov, *Faber–Krahn inequalities in sharp quantitative form*, Duke Math. J. **164** (2015), 1777–1831 (arXiv:1306.0392).